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Department of Artificial Intelligence and Machine Learning

FinSight — An Agentic AI Based Financial Research & Analysis System

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Problem Statement and Motivation

- **Financial research requires information from multiple sources** such as financial reports, market data, company filings, and news.
- **Manual financial analysis is time-consuming** and makes it difficult to efficiently combine and compare information from different sources.
- **Traditional LLM and basic RAG systems may provide incomplete or unsupported insights** because retrieval is generally fixed and does not adapt to the research requirement.
- **Need:** An intelligent financial research system that **plans research tasks, retrieves relevant evidence, analyzes financial information, validates its findings, and provides reliable insights with human oversight.**

User and Stakeholder Analysis

- **Financial Researchers / Analysts** – Quickly gather and analyze information about companies and markets.
- **Investors / Decision Makers** – Understand financial performance, opportunities, and potential risks.
- **Students / Researchers** – Perform structured financial research using multiple reliable sources.

Expected Outcome

- Generate **evidence-backed financial research reports** with financial insights, market information, risk analysis, and supporting sources.

Proposed agentic AI solution

FinSight is a **multi-agent AI system** that automates financial research and analysis.

- **Planner Agent** – Understands the user query and breaks it into research tasks.
- **Financial Research Agent** – Retrieves and analyzes financial reports, filings, and company information.
- **Market Data Agent** – Retrieves market data and performs relevant calculations.
- **News & Sentiment Agent** – Retrieves relevant news and analyzes market sentiment.
- **Risk Analysis Agent** – Identifies financial, market, competitive, and regulatory risks.
- **Synthesis Agent** – Combines the findings into a structured financial research report.
- **Critic Agent** – Verifies evidence and identifies unsupported or incomplete findings.

Dataset and knowledge source identification

- **Financial Reports & Company Filings**
  - Source: SEC EDGAR / Company Investor Relations
  - Annual reports, quarterly reports and financial statements
- **Market & Financial Data**
  - Source: Alpha Vantage APIs
  - Historical prices, volume and financial indicators
- **Knowledge Sources**
  - Company filings and financial reports
  - Regulatory disclosures
  - Financial news and industry reports
  - Company announcements

Selection of tools and Framework

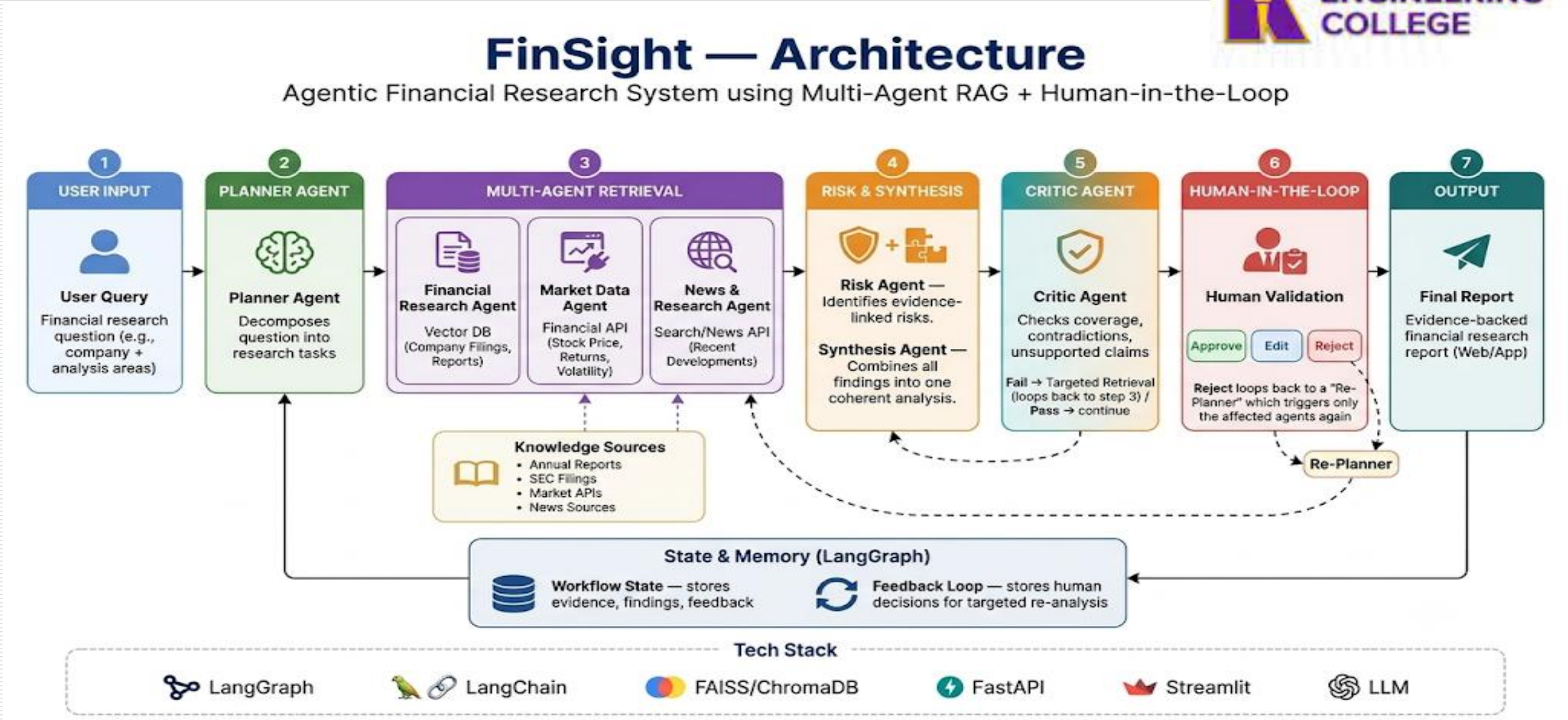
- **Database & Vector Store**
- SQLite / PostgreSQL – Stores user queries, workflow state, and report history
- ChromaDB – Stores and retrieves financial document embeddings
- **AI Model**
- GPT-4 / QWEN (pick per availability) – Generates agent reasoning and report synthesis
- **Agent Frameworks**
- LangChain – RAG, tool integration, retrievers
- LangGraph – Multi-agent workflow, state management, HITL checkpoints
- Sentence Transformers(all-MiniLM-L6-v2) – Text embeddings for document chunks

Selection of tools and Framework

- **Data & Tool Sources**
- SEC EDGAR / Company filings – Financial document ingestion
- Financial APIs (Alpha Vantage) – Market/stock data
- News/Search APIs – Recent developments and industry news
- **Deployment**
- Next.js 14 + React 18 + TypeScript + Tailwind CSS – Interactive research interface for user queries and reports
- FastAPI – Backend API and agent orchestration services
- **Visualization**
- Plotly – Interactive market charts and financial trends
- Matplotlib – Evaluation graphs (accuracy, hallucination rate, retrieval cost comparisons)

Architecture Diagram

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Agent workflow and Orchestration

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- **User Query** – User submits a financial research question (e.g., company + analysis areas)
- **Planning** – Planner Agent decomposes the question into research tasks and assigns agents
- **Multi-Agent Retrieval** – Financial, Market Data, and News Agents gather evidence in parallel
- **Risk Analysis** – Risk Agent identifies and evidence-links potential risk factors
- **Synthesis** – Synthesis Agent combines all agent outputs into one coherent analysis
- **Critic Validation** – Critic Agent checks coverage, contradictions, and unsupported claims
- **Human Validation** – User approves, rejects, or redirects specific findings

Agentic RAG

- **Task Identification** – Financial Agent determines what evidence is required (e.g., revenue, cash flow)
- **Query Generation** – Agent creates a targeted retrieval query for the vector DB
- **Vector Retrieval** – Fetches relevant chunks from company filings and reports
- **Evidence Evaluation** – Checks coverage, source quality, recency, and contradictions
- **Iterative Retrieval** – If evidence is insufficient, generates a refined query and retrieves again
- **Baseline Comparison** – Evaluates No-RAG, Basic RAG, and Agentic RAG on accuracy, hallucination rate, and retrieval cost

Human-in-the-Loop (HITL)

- **AI Findings** – System generates key findings with confidence levels (High/Medium/Low)
- **Human Review** – User inspects each finding along with its supporting evidence
- **Approve** – Accepts the finding as-is
- **Reject** – Flags a finding as outdated, wrong, or insufficiently supported, with feedback (e.g., "regulatory source is outdated")
- **Re-Planning** – Re-planner uses the feedback to trigger targeted retrieval on only the affected topic
- **Targeted Re-analysis** – Only the relevant agents (e.g., News → Risk → Synthesis) re-run, not the full pipeline
- **Final Validation** – Human re-reviews the updated finding before the final report is generated

SYSTEM IMPLEMENTATION

Functional Prototype

- **Research Dashboard** – Query formulation and company selection
- **Multi-Agent Execution** – Planning → SEC / Market / News → Synthesis → Verification → HITL
- **Evidence Trace** – SEC chunks, citations and source provenance
- **HITL Audit Console** – Approve, Edit and Reject findings
- **Executive Dashboard** – Verified KPIs and financial trend charts
- **Final Report** – Audited findings, evidence limitations and PDF export

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Source-Code Repository

- **Frontend:** Next.js 14, React, Tailwind CSS
- **Backend:** FastAPI, LangGraph
- **Agents:** Planner, Financial Research, Market Data, News, Risk, Synthesis, Verifier, Critic, Replanner
- **Data:** ChromaDB + SEC filings + Alpha Vantage
- **Evaluation:** 49 automated tests + benchmark suite

EXPERIMENTAL EVALUATION AND RESULTS

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Evaluation Metrics

Metric	FinSight Result
Core regression test pass rate	100% (23/23)
Expanded automated test pass rate	100% (49/49)
Routing intent accuracy	100% (5/5)
News relevance test success	100% (6/6)
Numerical verification tests	100% (12/12)
Finsight Citation accuracy	90.0%
Basic RAG citation accuracy	67.5%
Finsight Hallucination rate	10.0%
Basic RAG hallucination rate	32.5%
No-RAG hallucination rate	42.5%

Testing Methodology

Query → Intent Routing → RAG / Market / News → Verification → Critic → HITL → Final Report

- Unit & integration testing
- Routing and retrieval testing

- News relevance testing
- Numerical verification

Performance

- Apple: **~22.4 s**
- NVIDIA: **~24.1 s**
- Microsoft: **~21.8 s**
- **13 LLM calls/run**

Key Result:

FinSight achieved **90% citation accuracy** and reduced hallucination to **10%**, compared with **67.5% / 32.5%** for Basic RAG.

COMPARISON

COMPARISON WITH BASELINES

Capability	No-RAG	Basic RAG	FinSight
Evidence Grounding	❌	✓	✓ Agentic
Query Decomposition	❌	Limited	✓
Numerical Verification	❌	Limited	✓
News Filtering	❌	Basic	✓
Quality Control	❌	Basic	✓ Fail-Closed
Human Review	❌	❌	✓ HITL

INNOVATION

CORE INNOVATIONS

1. Deterministic Causal Attribution

Separates verified financial figures from unsupported causes

2. Multi-Tier News Relevance Filtering

Rejects irrelevant portfolio/ticker noise

3. Fail-Closed Coverage Ledger

Reports evidence limitations instead of fabricating missing data

4. Human-in-the-Loop Revalidation

Analyst edits are re-verified before publication

CONCLUSION AND FUTUREWORK

- **Conclusion**
- FinSight combines agentic research, evidence retrieval, deterministic verification and human oversight to produce **auditable and evidence-grounded financial research**.

Future Work

- Expand financial document corpus
- Integrate live SEC data
- Improve news-source coverage
- Parallelize research tasks
- Cloud deployment & multi-user support

Key Takeaway:

From Query → Evidence → Verification → Human Audit → Trusted Research Report

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Thank You